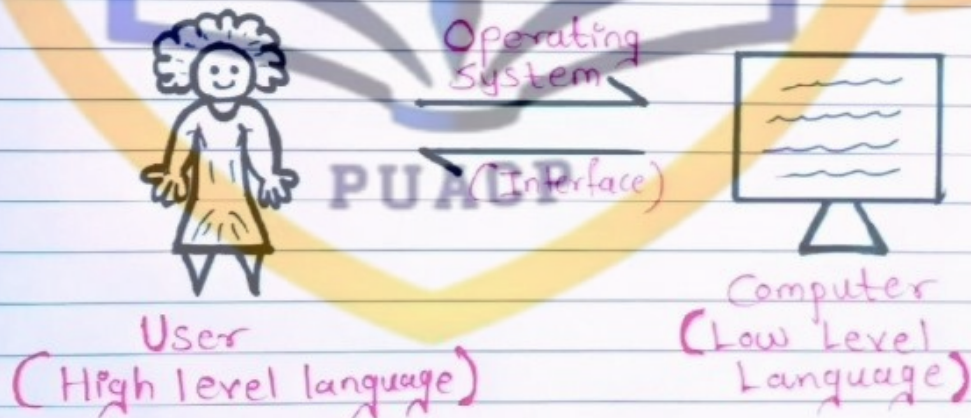


Operating System

* What is operating system ?

⇒ An Operating System (OS) is an interface between a computer user and computer hardware. An Operating System is a software which performs all the basic tasks like file management, memory management, process management, handling input and output, & controlling peripheral devices such as disk drives and printers.

For Example :-



- Operating System is a collection of set of programs which manages all the resources of the computer system. It is an intermediate between user & computer and the computer hardware.

Types of Operating System

1. Batch Operating System
2. Distributed Operating System
3. Network Operating System
4. Time Sharing Operating System
5. Real Time Operating System
6. Multiprocessing Operating System
7. Embedded Operating System
8. Multiprogramming Operating System

1. Batch Operating System

Batch operating systems process data in the form of batches. The user of a batch operating system never directly interacts with the computer. In this type of OS, every user prepares his or her job on an offline device like a punch card and submit it to the computer operator.

Batch operating system was the first operating system to be used in second generation computer.

2. Distributed Operating System

Failure of one system will not affect the other systems because all the computers are independent of each other.

The load on the host system is reduced. With resource sharing facility, a user at one site may be able to use the resources available at another.

Failure of the main network will lead to the failure of the whole system.

The setup cost is high. Software used for such systems is highly complex.

3. Network Operating System

Popular Network Operating Systems:-

Windows NT

Window Server 2003

Window Server 2008

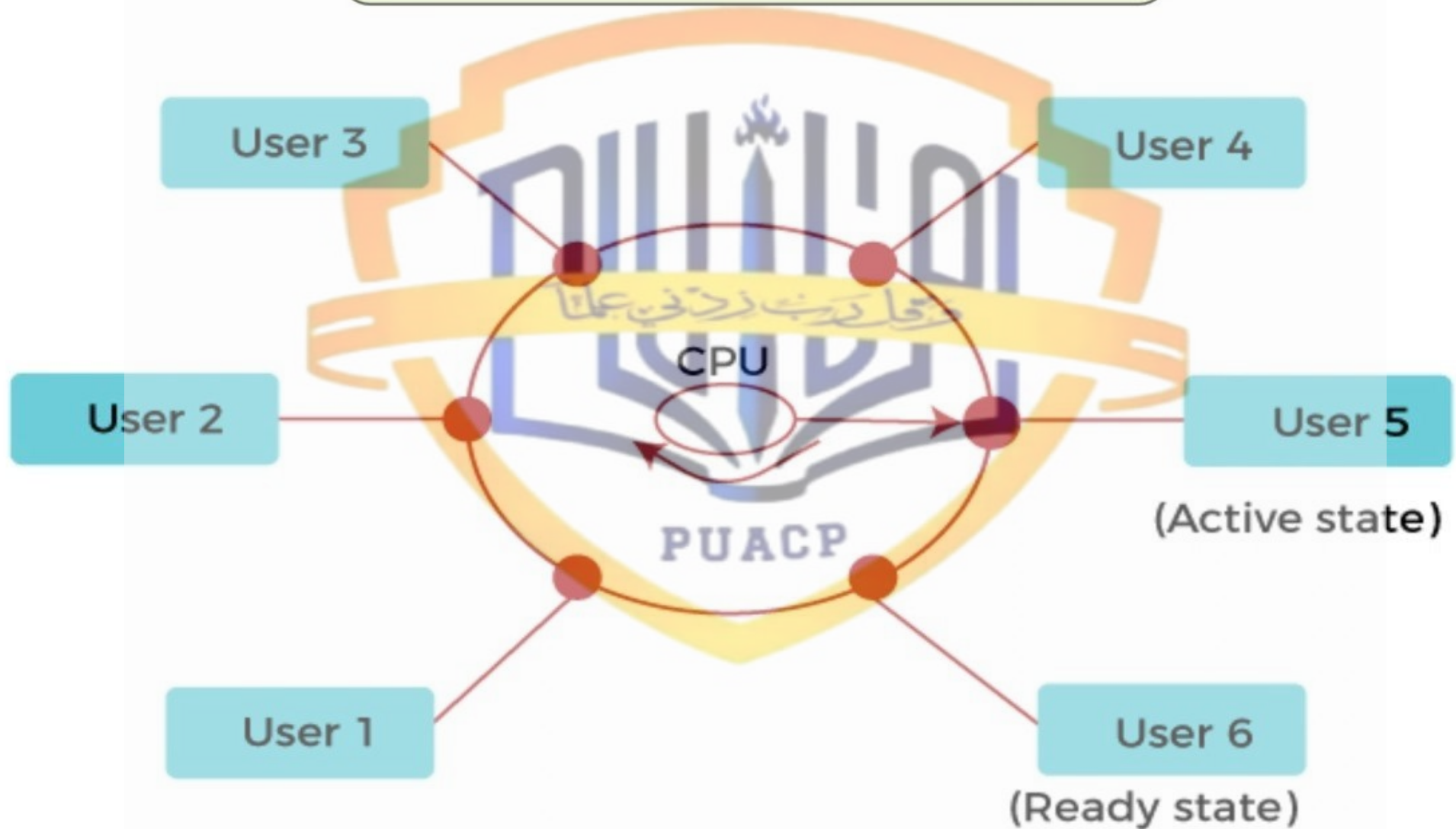
Linux

Unix

Mac OS



4. Time Sharing Operating System



4. Time Sharing Operating System

Each task gets equal time for execution. The time allotted to execute one task is called a quantum, and as soon as the time to execute one task is completed, the system switches over to another task.

5. Real Time Operating System

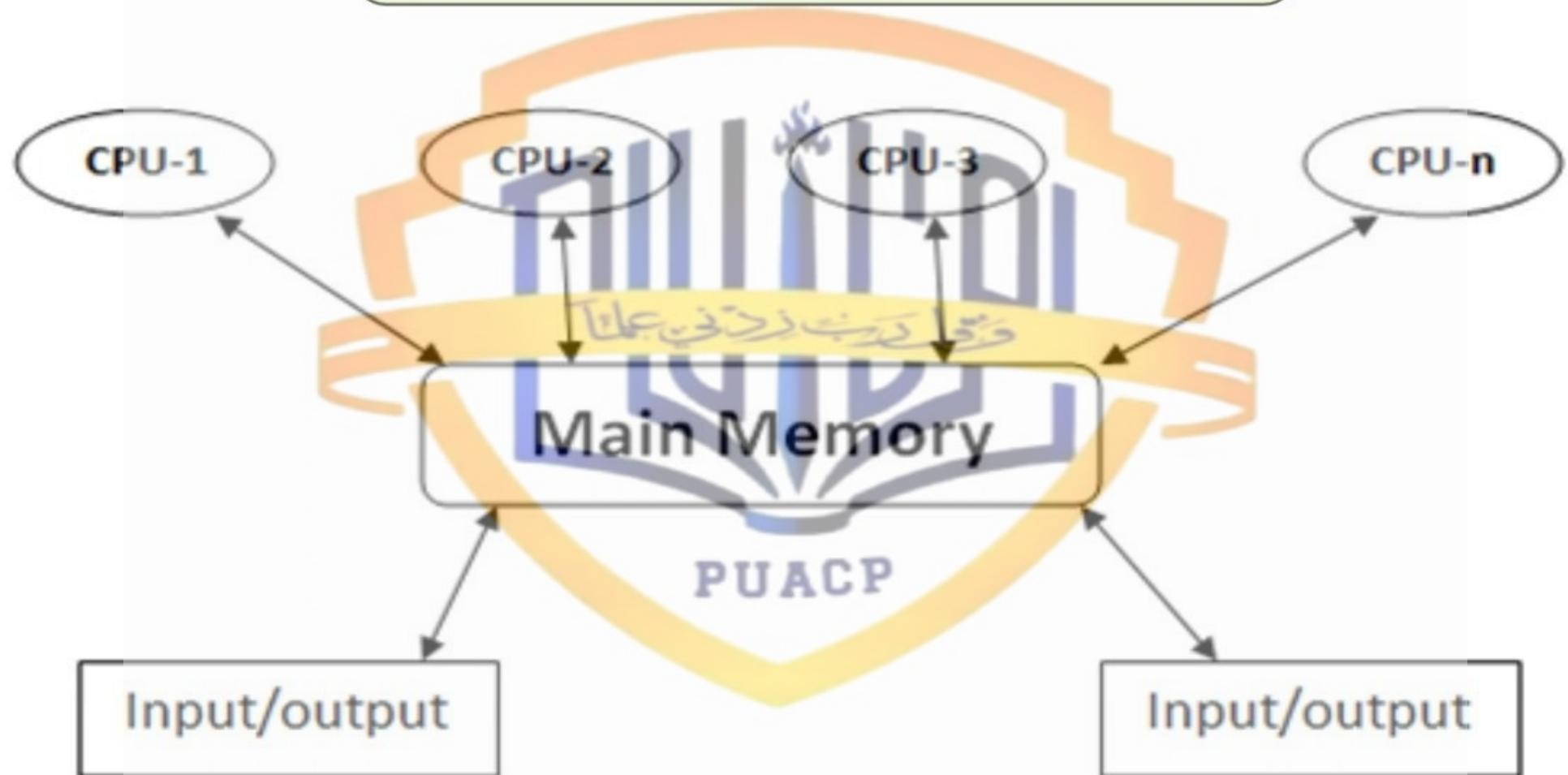
Real Time Operating System is an advanced operating system and serve real-time systems. The response time between input, processing and response is tiny, which is beneficial for processes that are highly sensitive and need high precision. These operating systems are useful when many events occur in a short time or within certain deadlines, such as real-time simulations.

5. Real Time Operating System

1. **Hard Real-time system** :- Hard Real Time OS doesn't accept any delay even the shortest possible delay in the processing time. Here the time constraint is very strict.

2. **Soft Real-time system**:- The soft real-time OS is the operating system for applications where time constraint is not very strict.

6. Multiprocessing Operating System



7. Embedded Operating System

An embedded operating system is a computer operating system designed for use in embedded computer systems. It has limited features. The main goal of designing an embedded operating system is to perform specified tasks for non-computer devices.

Computer: *autosaves file*

Me: Aww thanks! Where is it?

Computer:



I don't know, man.